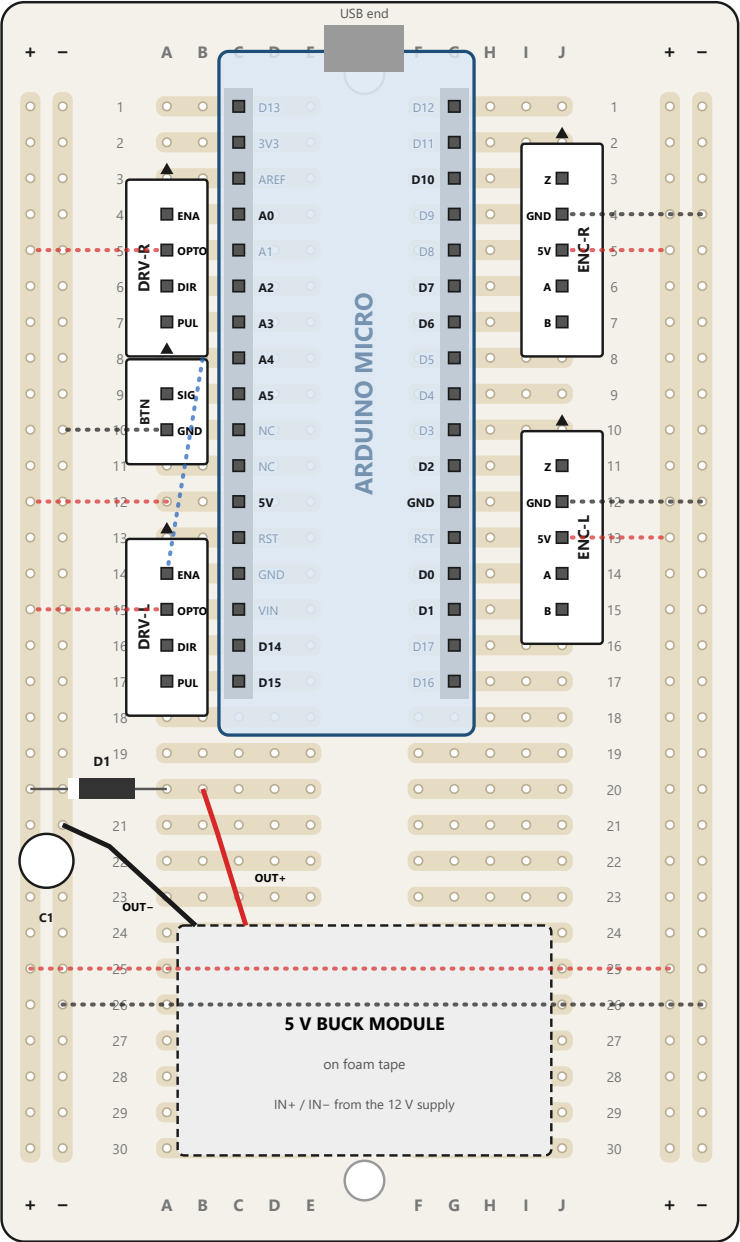


Triangle carrier board: component side



Placement (board coordinates as printed)

- Micro** 2 × 17-way female header sockets in **C1–C17** and **G1–G17**, USB end at row 1. Fit the top centre mounting screw first: it sits under the Micro.
- ENC-R** JST-XH 5-pin in **J3–J7**, pin 1 at J3. Right motor encoder.
- ENC-L** JST-XH 5-pin in **J11–J15**, pin 1 at J11. Left motor encoder.
- DRV-R** JST-XH 4-pin in **A4–A7**, pin 1 at A4. Right driver.
- DRV-L** JST-XH 4-pin in **A14–A17**, pin 1 at A14. Left driver.
- BTN** JST-XH 2-pin in **A9–A10**, pin 1 at A9. Start button (optional).
- D1** 1N5819: anode in **A20**, cathode (band) in the **left + rail** by row 20.
- C1** 100 µF 16 V across the left rails by row 22: + leg in the + rail.
- Buck** 5 V buck module on foam tape in the empty area (rows 24–30). OUT+ lead to **B20**, OUT– lead to the **left – rail** by row 21. IN+ / IN– from the 12 V supply (own pair).

Cables

Pin	ENC-R / ENC-L	DRV-R / DRV-L	BTN
1	YEL Z	ENA	button
2	BLK GND	OPTO	button
3	RED 5V	DIR	
4	BRN A	PUL	
5	BLU B		

- **The encoder pin order has changed** from the prototype: crimp to this table. Cut back ORG, GRN and WHT and insulate each.
- Driver cables follow the DM320T’s P1 order. Motor leads go straight to the drivers.

How it works

- Each connector sits on the same strips as the Micro pins it uses, so **no signal needs a wire**: the copper joins them.
- Where a connector needs 5 V or GND instead, that strip is **cut** between the connector and the Micro, and a short link on the underside joins the connector pin to a rail (page 2).
- The Micro pins on cut strips (A1, VIN, the left GND, D9, D8, RST) are left unused.

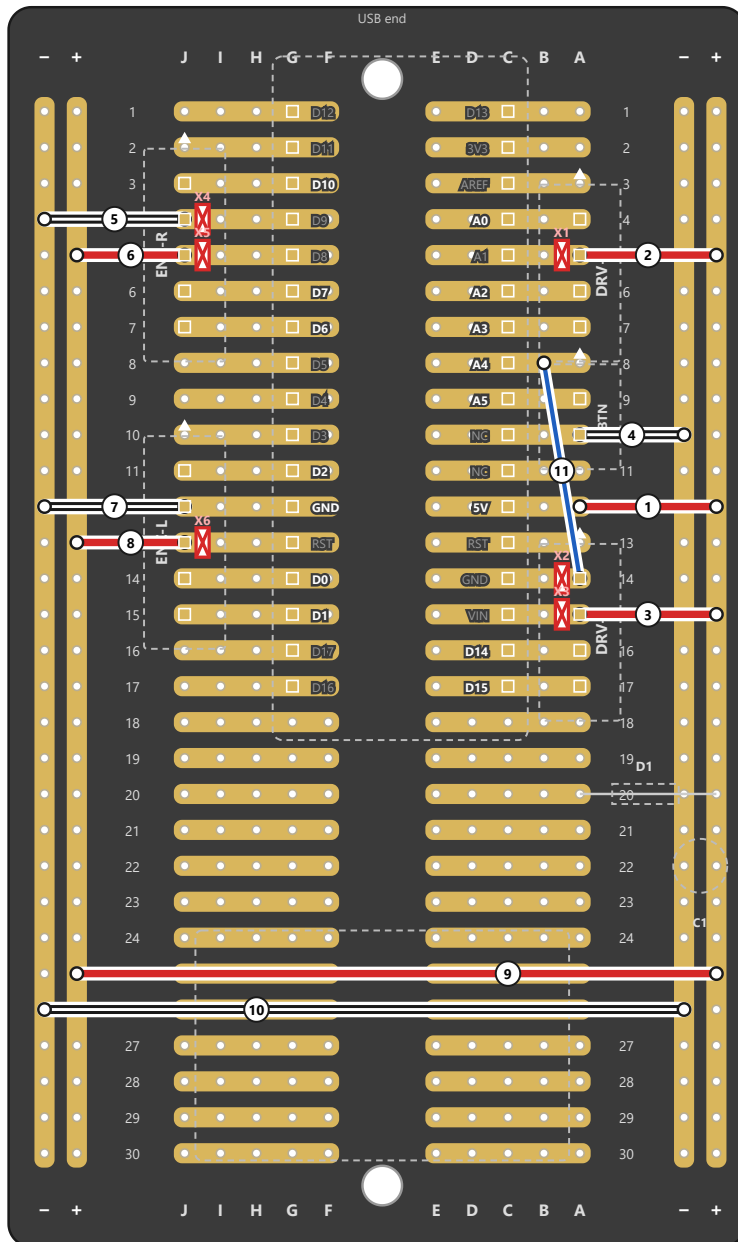
Connector	Pins reach (top to bottom)
ENC-R	D10, GND*, 5V*, D7, D6
ENC-L	D2, GND, 5V*, D0, D1
DRV-R	A0, 5V*, A2, A3
DRV-L	A4 (link 11)*, 5V*, D14, D15
BTN	A5, GND*

* through a cut and an underside link

- Header or socket pin
- Link on the underside (page 2)

Underside: track cuts and links

Board turned over left to right: the connectors in column A are now on the **right**



Track cuts

#	Between	Isolates
☐ X1	A5 – B5	DRV-R OPTO from Micro A1
☐ X2	A14 – B14	DRV-L ENA from Micro GND
☐ X3	A15 – B15	DRV-L OPTO from Micro VIN
☐ X4	I4 – J4	ENC-R GND from Micro D9
☐ X5	I5 – J5	ENC-R 5V from Micro D8
☐ X6	I13 – J13	ENC-L 5V from Micro RST

Links (all on the underside)

#	From	To	What
☐ 1	A12	left + rail, row 12	Micro 5V (C12)
☐ 2	A5	left + rail, row 5	DRV-R OPTO
☐ 3	A15	left + rail, row 15	DRV-L OPTO
☐ 4	A10	left – rail, row 10	BTN GND (row 10 is an unused NC pin)
☐ 5	J4	right – rail, row 4	ENC-R GND
☐ 6	J5	right + rail, row 5	ENC-R 5V
☐ 7	J12	right – rail, row 12	Micro GND to the right – rail
☐ 8	J13	right + rail, row 13	ENC-L 5V
☐ 9	left + rail, row 25	right + rail, row 25	Left + rail to right + rail
☐ 10	left – rail, row 26	right – rail, row 26	Left – rail to right – rail
☐ 11	A14	B8	DRV-L ENA to Micro A4

Build and test

- **Before anything:** check with a meter that no + rail is joined to a – rail.
- **Cuts first,** with a sharp knife or a 3 mm drill bit twisted by hand. Check with a meter that each one is open.
- **Then:** mounting screw under the Micro, sockets (plug the Micro in to hold them straight while soldering), XH headers, D1, C1, buck leads, links 1–11.
- **Before fitting the Micro:** check the + rails to the – rails isn't a short. Connect 12 V to the buck: both + rails should read about 4.6–5.1 V (set an adjustable buck to 5.3 V). Check each connector's 5V and GND pins.
- **With the Micro fitted:** set `CARRIER_BOARD true` in `config.h`, run the test sketch and use `t` to check each PUL and DIR at the driver end.
- **D1** stops the buck and a USB lead fighting, so USB can stay plugged in with the 12 V on.

-  Track cut (X1-X6)
-  5 V link (thick)
-  GND link (hollow)
-  Signal link (thin)
-  Solder the link here